CSTA San Diego

AP Computer Science: Data-Driven Planning

Leveraging student assessment data for impactful instruction.

UC San Diego
CReaTe
Center for Research on Educational Equity, Assessment & Teaching Excellence

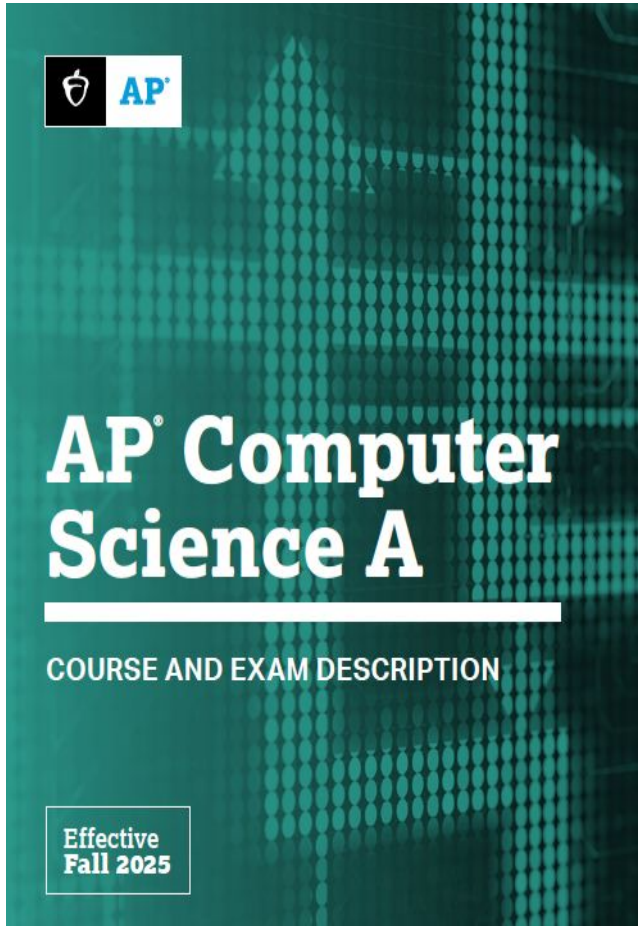
san diego county office of EDUCATION
FUTURE WITHOUT BOUNDARIES℠

Setting the Stage

Defining the Tracks

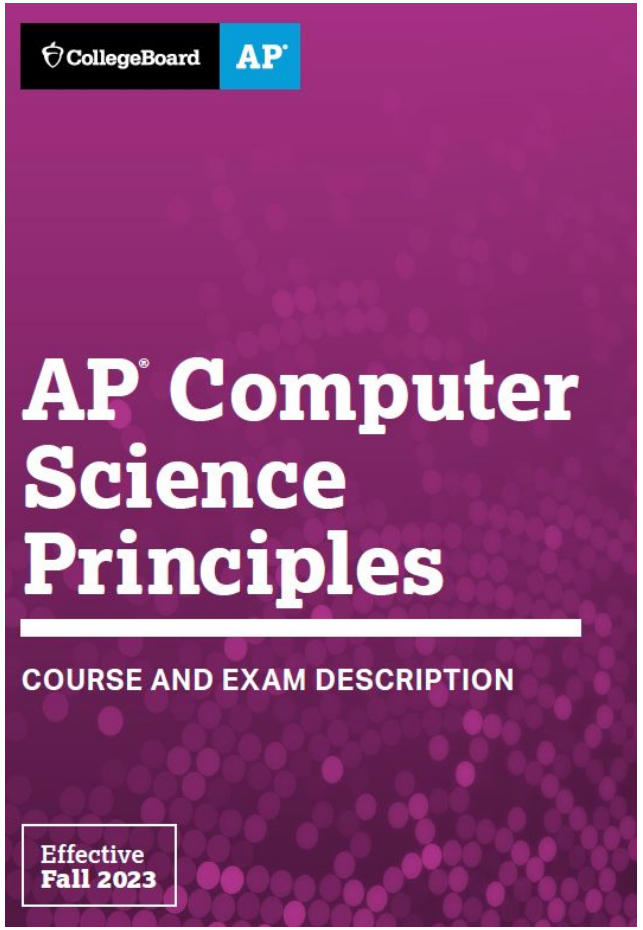
AP CS A

Technical, object-oriented focus using Java. Rigorous study of programming fundamentals and algorithms.



AP CS Principles

Breadth-first, societal-impact focus. Broad introduction to computing, data science, and internet systems.



Shared Goals & Data

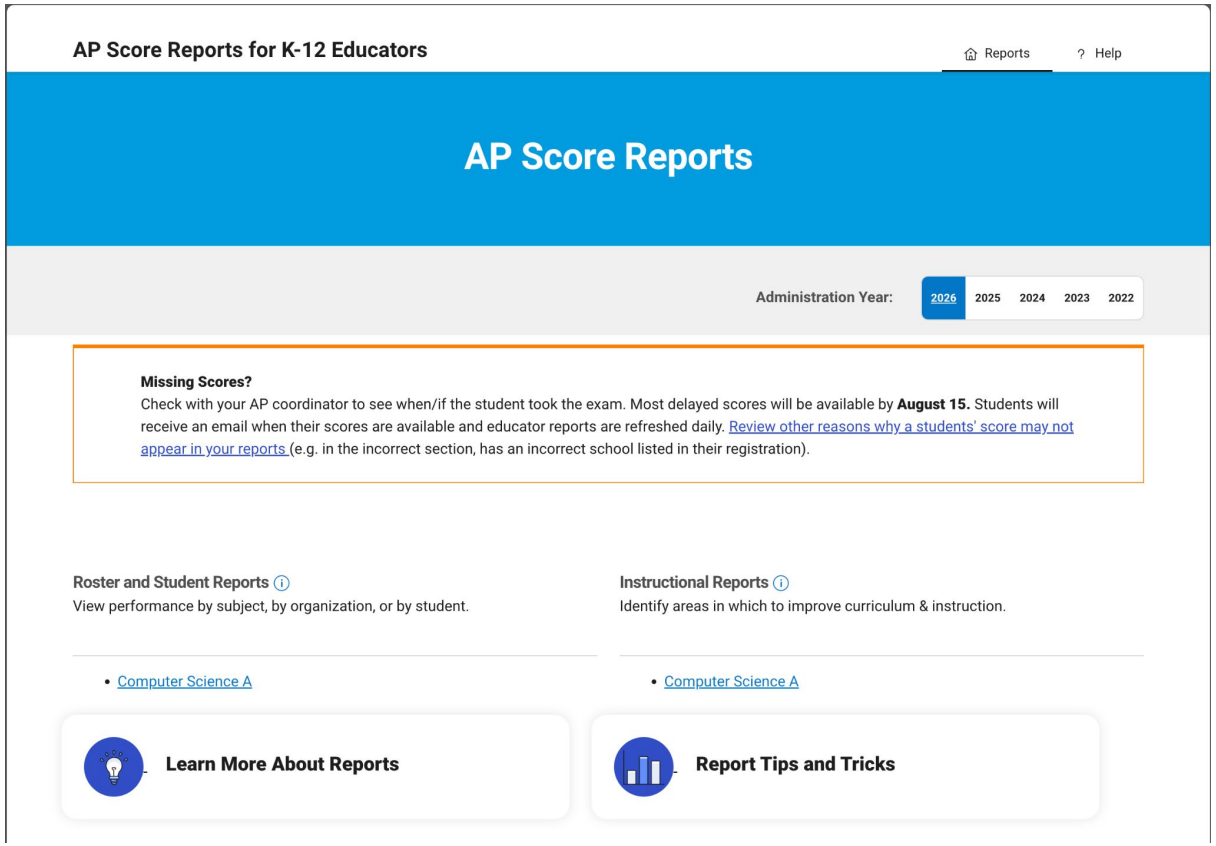
- **Broadening Participation:** Both courses aim to prepare students for a digital-first future.
- **Data is a Tool:** It is not about judging teaching, but mapping student conceptual mastery.
- **Informed Planning:** Identifying specific areas for growth across distinct pathways.

Navigating Your Data

Accessing Your AP Reports

A step-by-step guide to locating and downloading your score rosters and instructional planning data.

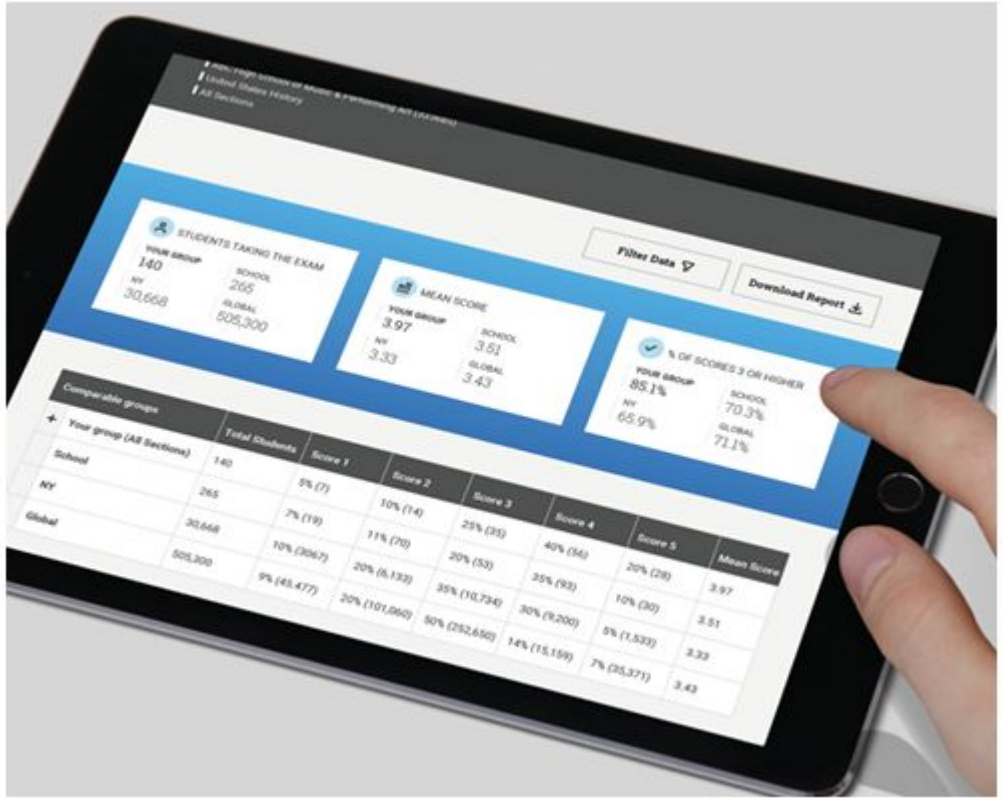
Go to scores.collegeboard.org and log in with your educator account to access the score reporting portal.



Roster and Student Reports

Use the Roster to:

- Compare classroom performance vs. state and global peers.
- Benchmark performance without relying on a single metric.
- Identify aggregate trends across students.



Instructional Reports

How to read the report:

- **Drill Down:** Focus on specific topic performance units.
- **Look for Ratios:** Compare group means to global means.
- **Actionable Insight:** Use the filter to target specific scoring groups (e.g., scores of 2s vs 3s).



Gaps: Data vs. Instruction

Data Gaps

Where are students scoring lower than expected based on benchmarks?

Instructional Gaps

Are they struggling with logic (algorithms) or representation (syntax)?

Course Specifics

AP CSA

MCQ Report: Includes 4 units and 5 practices categories, though some may will not have data.

FRQ Analysis: Compare group means per point to state and global averages. Focus on question totals since point descriptions are currently pending.

AP CSP

MCQ Report: Includes 5 big ideas and 5 practices and 3 question type categories,, though some sections may not have data.

Performance Tasks: Compare group means for the video, the requirements and the written responses to state and global averages.

Forms

Students take different versions of the test called forms. Groups are built around forms since different versions of the test result in different median scores. We will ignore this at our own risk later.

Data Aggregation Tool

Simplify your view of your data. https://scerruti.github.io/AP-Score-Reports/

Download your reports and process them with this tool to create a single aggregated data report.

Data never leaves your machine.

Aggregation is not strictly statistically accurate but it should be adequate for our purposes.

Before we dive into the data....

Using AP Classroom to be data driven during the year...



1. High-Level Reports

Instructional planning reports offer retrospective data that is often too aggregated or high-level to guide day-to-day, real-time adjustments in the classroom.



2. AP Classroom Solution

By utilizing AP Classroom for ongoing assessments throughout the year, you unlock detailed, granular student insights to actively support and drive learning.

Individual Review & Planning

Review your student scores individually and prepare your strategy using the provided planning handout.

1. Review Student Scores

Analyze overall and individual performance metrics to identify gaps in understanding.

- Look for computational thinking trends
- Pinpoint specific low-scoring coding tasks
- Compare benchmark trends across classes

2. Plan for the Deep Dive

Formulate target questions and concrete goals to guide your upcoming collaborative session.

- **Use the provided handout**
- Document specific intervention strategies
- Draft focus areas for peer feedback

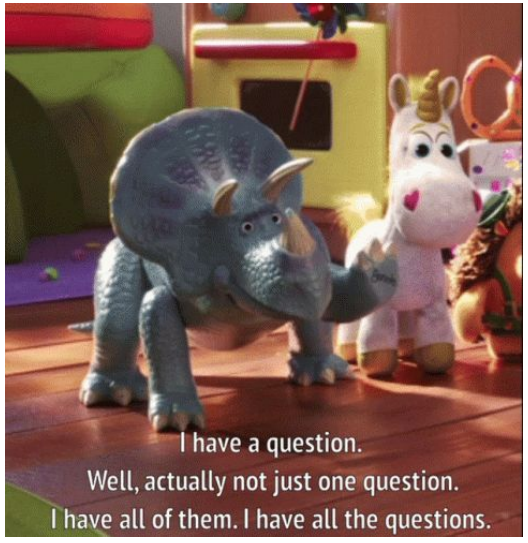


Questions?

Let's unpack your data in the Deep Dive session.







Data Deep Dive







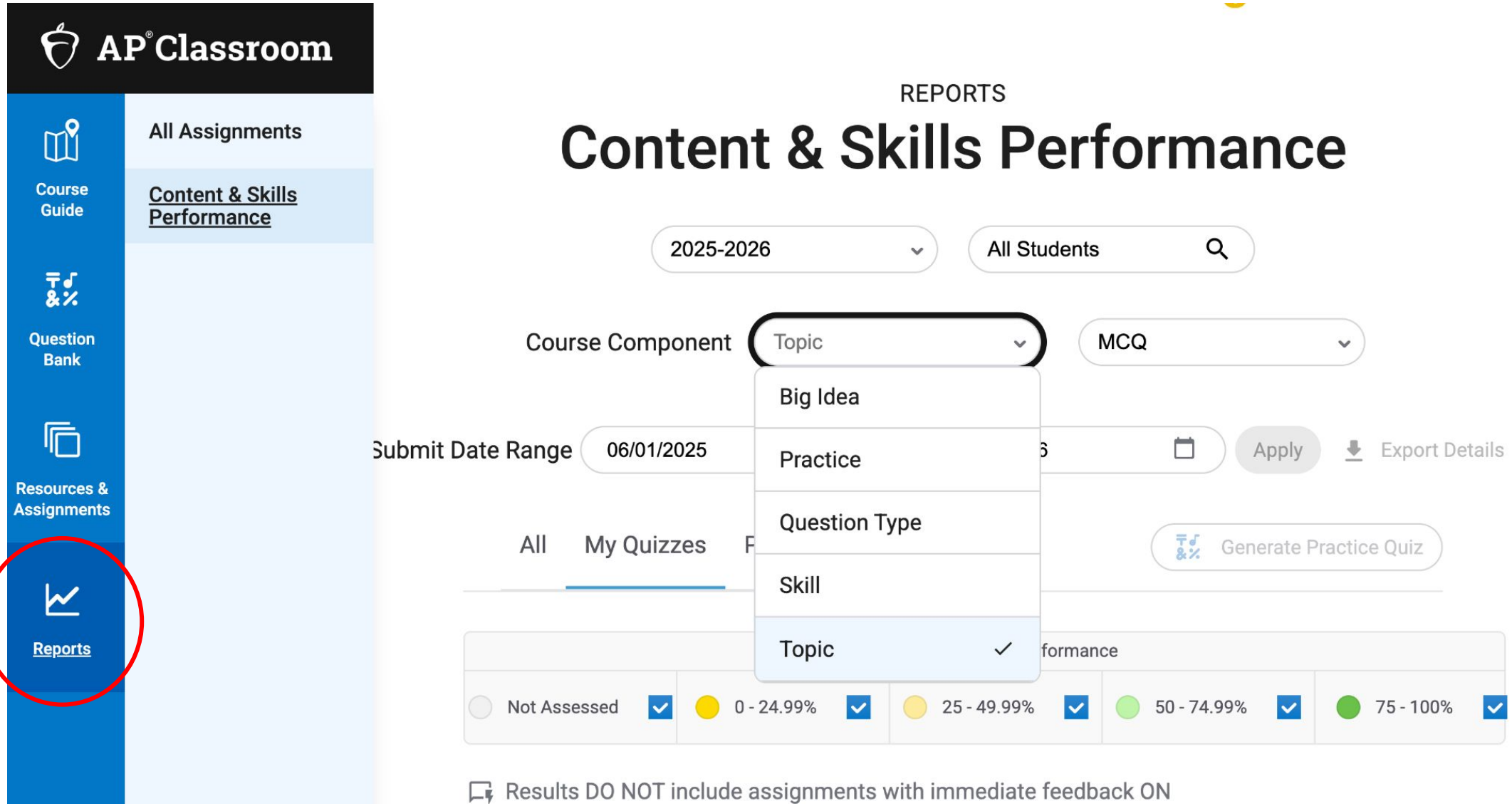
Lunch!

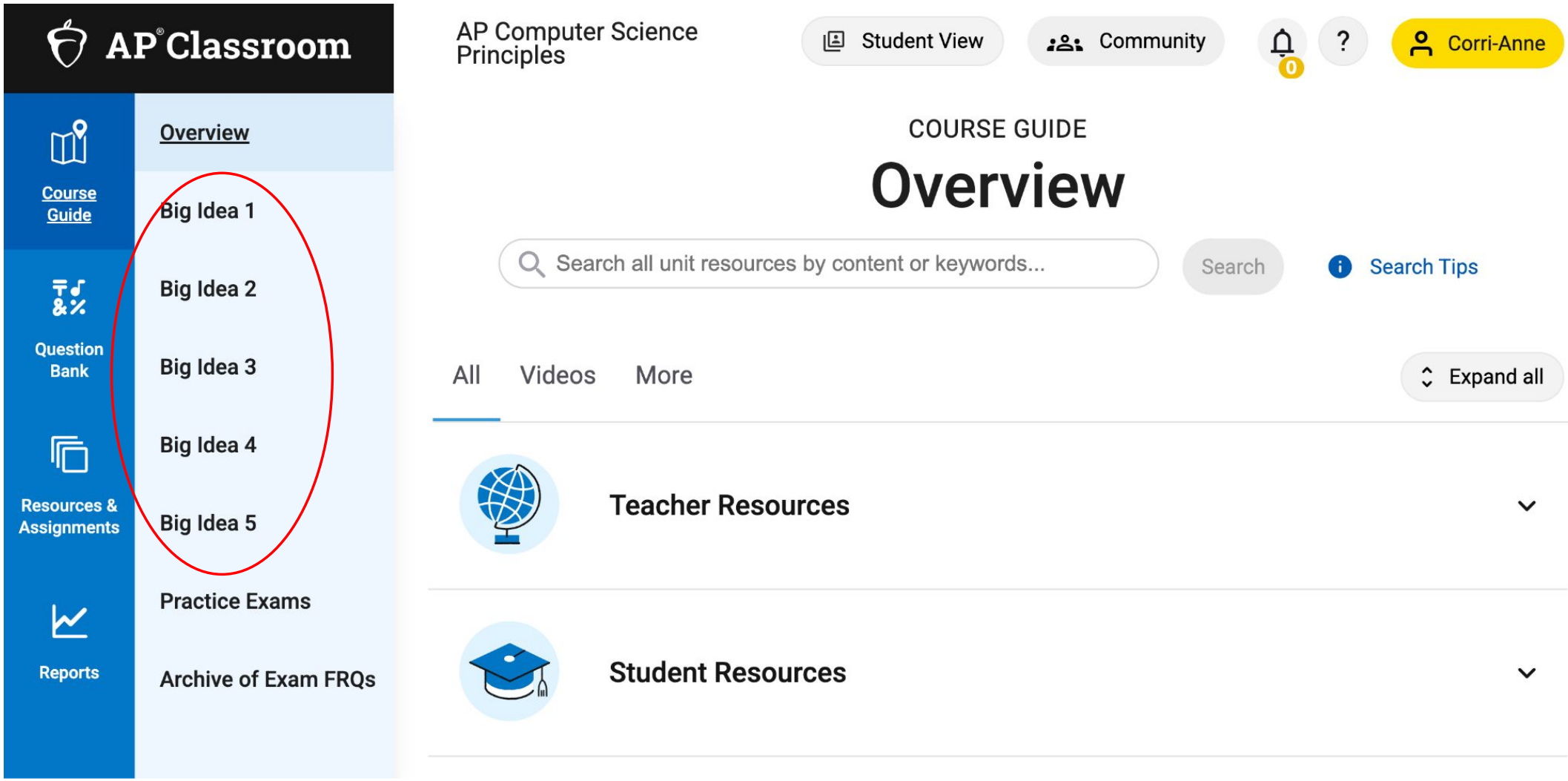
Is an excellent time for networking!

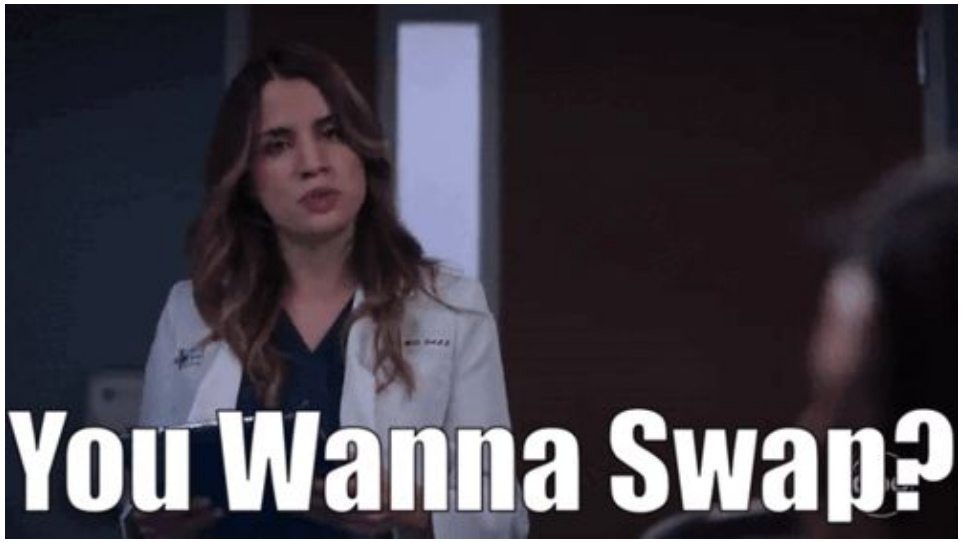


College Board - reflection and planning

Compare class assessments to exam



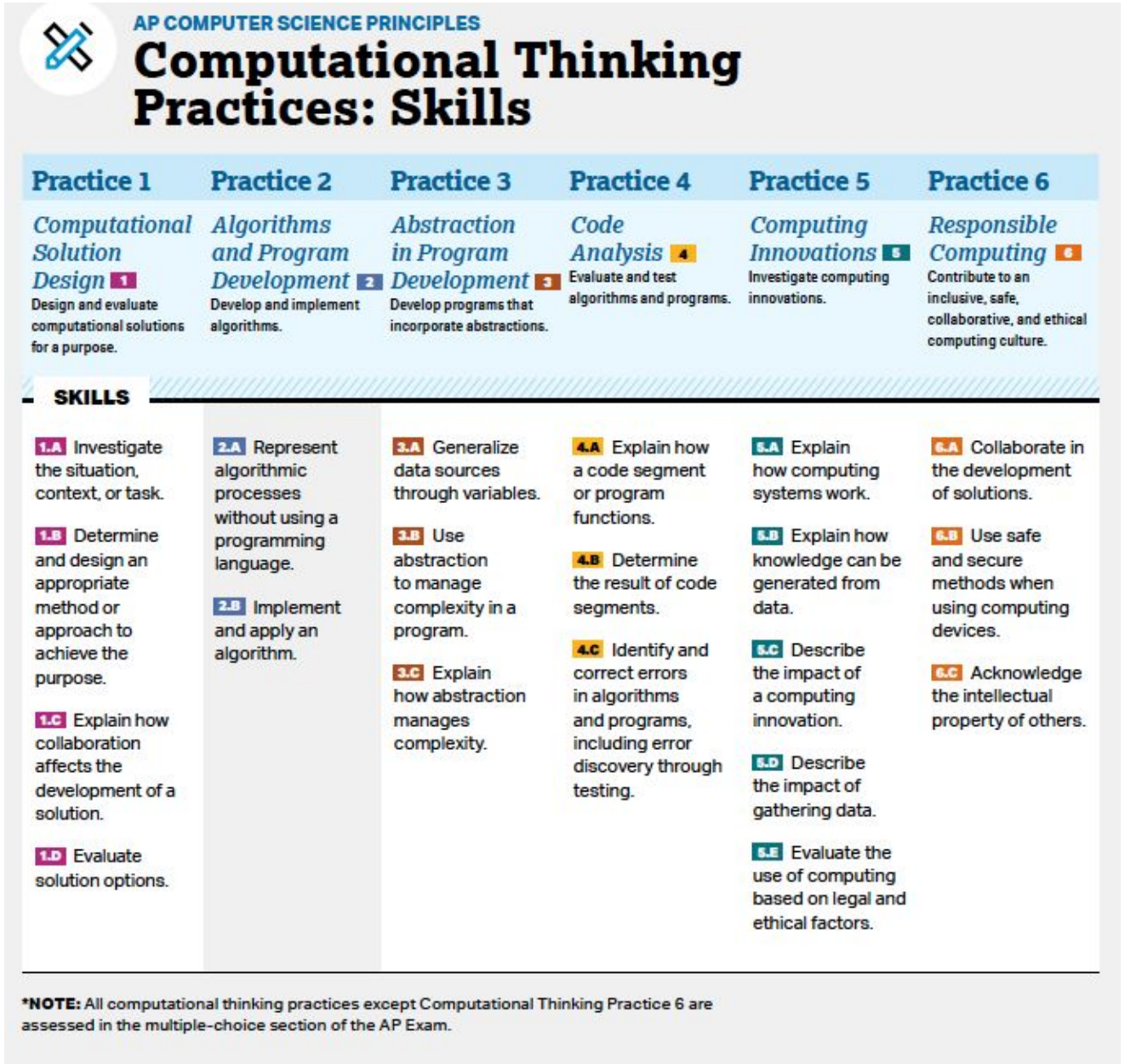

AP® Classroom
AP Computer Science Principles
Student View
Community
0
Corri-Anne
Course Guide
Overview
Big Idea 1
Big Idea 2
Big Idea 3
Big Idea 4
Big Idea 5
Question Bank
Resources & Assignments
Practice Exams
Reports
Archive of Exam FRQs
COURSE GUIDE
Overview
Search all unit resources by content or keywords...
Search
Search Tips
All
Videos
More
Expand all
Teacher Resources
Student Resources





Strategy Swap Roundtable

Roundtable Swap



AP COMPUTER SCIENCE PRINCIPLES

Computational Thinking Practices: Skills

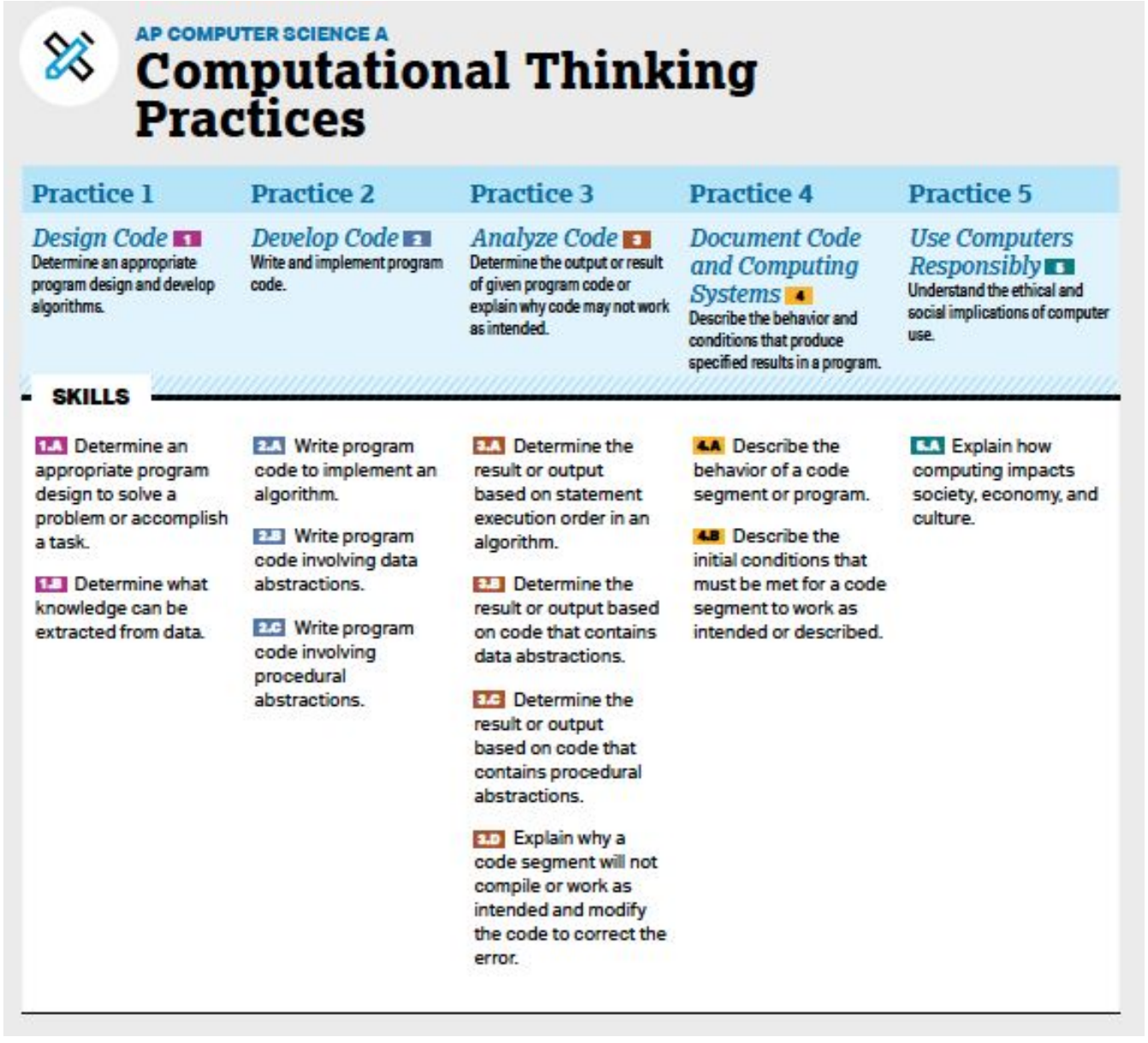
Practice 1	Practice 2	Practice 3	Practice 4	Practice 5	Practice 6
Computational Solution Design 1 Design and evaluate computational solutions for a purpose.	*Algorithms and Program Development* 2 Develop and implement algorithms.	*Abstraction in Program Development* 3 Develop programs that incorporate abstractions.	*Code Analysis* 4 Evaluate and test algorithms and programs.	*Computing Innovations* 5 Investigate computing innovations.	*Responsible Computing* 6 Contribute to an inclusive, safe, collaborative, and ethical computing culture.
SKILLS					
1.A Investigate the situation, context, or task.	2.A Represent algorithmic processes without using a programming language.	3.A Generalize data sources through variables.	4.A Explain how a code segment or program functions.	5.A Explain how computing systems work.	6.A Collaborate in the development of solutions.
1.B Determine and design an appropriate method or approach to achieve the purpose.	2.B Implement and apply an algorithm.	3.B Use abstraction to manage complexity in a program.	4.B Determine the result of code segments.	5.B Explain how knowledge can be generated from data.	6.B Use safe and secure methods when using computing devices.
1.C Explain how collaboration affects the development of a solution.		3.C Explain how abstraction manages complexity.	4.C Identify and correct errors in algorithms and programs, including error discovery through testing.	5.C Describe the impact of a computing innovation.	6.C Acknowledge the intellectual property of others.
1.D Evaluate solution options.				5.D Describe the impact of gathering data.	
				5.E Evaluate the use of computing based on legal and ethical factors.	

***NOTE:** All computational thinking practices except Computational Thinking Practice 6 are assessed in the multiple-choice section of the AP Exam.

For AP CSP :

https://tinyurl.com/APCoPSDPrinciples

Roundtable Swap



AP COMPUTER SCIENCE A

Computational Thinking Practices

Practice 1	Practice 2	Practice 3	Practice 4	Practice 5
Design Code 1 Determine an appropriate program design and develop algorithms.	*Develop Code* 2 Write and implement program code.	*Analyze Code* 3 Determine the output or result of given program code or explain why code may not work as intended.	*Document Code and Computing Systems* 4 Describe the behavior and conditions that produce specified results in a program.	*Use Computers Responsibly* 5 Understand the ethical and social implications of computer use.

SKILLS

1.A Determine an appropriate program design to solve a problem or accomplish a task. 1.B Determine what knowledge can be extracted from data.	2.A Write program code to implement an algorithm. 2.B Write program code involving data abstractions. 2.C Write program code involving procedural abstractions.	3.A Determine the result or output based on statement execution order in an algorithm. 3.B Determine the result or output based on code that contains data abstractions. 3.C Determine the result or output based on code that contains procedural abstractions. 3.D Explain why a code segment will not compile or work as intended and modify the code to correct the error.	4.A Describe the behavior of a code segment or program. 4.B Describe the initial conditions that must be met for a code segment to work as intended or described.	5.A Explain how computing impacts society, economy, and culture.

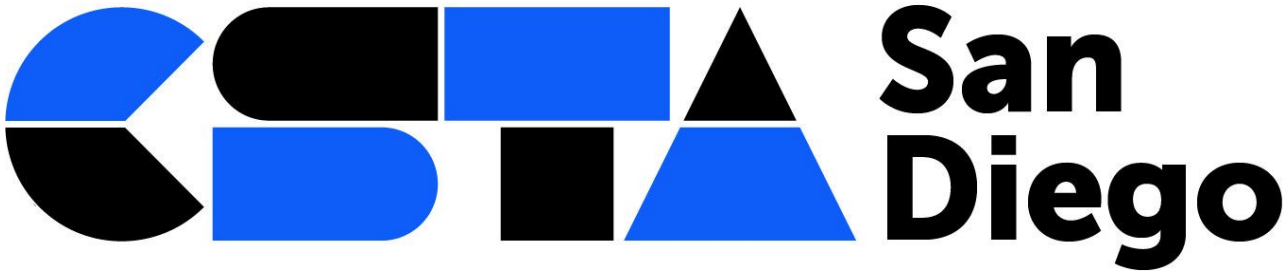
For AP CSA : https://tinyurl.com/APCoPSDA

Closing Moves

We truly hope you got a lot out of today's Session!

A few further considerations. . .

Part of the world's largest Computer Science Department!



Become a full and active member of our community!

- Become a full member by paying $50 dues per year. There is a discount code that can make your first year only $10!
- Benefits:
  - Invitation to Professional Learning Events and Chapter Meetings
  - Networking with other CS Teachers locally and internationally
  - Free online Professional Learning (PD)
  - Voting Rights
  - Discounts for the National Conference registration
  - Leadership opportunities.
  - Newsletters and publication subscriptions

CSTA San Diego



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Let's Keep the Conversations Going!

Announcing. . .

The AP CS Teacher Community of Practice

- Meets weekly to have conversations about teaching CS
- Build community with other AP CS teachers
- Share best practices
- Talk through challenges
- Think through new ideas together
- First meeting is planned for the third week in August (tentatively - see next slide)
- Look out for a personal invitation or a general announcement coming soon!

San Diego

We are always looking for what you're needs are!

Help us learn about you



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Take our experience survey to guide our thinking!

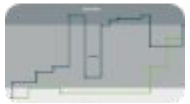
- Find the survey at https://tinyurl.com/CSTASDExperienceSurvey26

Thanks for joining us today!
We hope you have (had) a great start to the school year!



CSTA San Diego

Image Sources



https://nightingaledvs.com/wp-content/uploads/2021/08/Fig01and04.png

Source: nightingaledvs.com



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Source: www.vecteezy.com